

Propensity-Weighted Modified Logistic Regression for Robust PU Learning under Structural Label Noise

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Abstract. Positive-Unlabeled (PU) learning addresses classification when only a subset of positive samples is labeled and the remaining data are unlabeled. In realistic applications such as medical diagnosis, labeling is often feature-dependent rather than completely random, which produces structural noise under the Selected At Random (SAR) assumption. This paper constructs controlled random and structural PU noise mechanisms and proposes Propensity-weighted Modified Logistic Regression (P-MLR), which separates the target class distribution from feature-dependent labeling propensity through a parameterized weighting branch. Experiments on five tabular and non-tabular benchmarks show that P-MLR remains robust under severe structural bias, especially on medical and structured tabular data, while preserving linear-model interpretability and $\theta(d)$ online inference complexity.

Keywords: PU learning, structural noise, propensity score, linear model, robustness

1 Introduction

Positive-Unlabeled (PU) learning is an essential machine learning paradigm for classification modeling under conditions where only a limited number of positive samples and a massive amount of unlabeled samples are available, and is of profound significance in fields such as medical diagnosis, fraud detection, and recommendation systems, where obtaining reliable negative samples is often difficult while unlabeled data typically contains both positive and negative instances [1, 2]. Existing studies often assume that labeling noise is purely random, meaning the probability of a positive sample being labeled is constant and independent of its features [3]; however, in real-world application scenarios, this probability is frequently correlated with sample features—for example, patients with more obvious symptoms are more likely to be diagnosed—forming structured noise with structural dependencies that better reflects the actual data generation process [4, 5].

To overcome the challenge of unlabeled data in PU learning, traditional methodologies include two-step heuristic algorithms such as TRNS, which filter “reliable negative samples” from unlabeled data, and global distribution correction